LOCUS — Data Flow Specification

LEVEL 0: CONTEXT OVERVIEW & EXTERNAL ENTITY INTERFACES

Document: LOCUS-DFD-00
Standard: Gane & Sarson

The Level 0 Context Diagram abstracts the entire **LOCUS** (Living space and Occupancy Control Unified System) platform as a single operational process (**0.0**). In strict adherence to standard structured systems analysis, **no internal databases/data stores are exposed at this level**. All data flows represent direct transactions across four external boundary entities: Resident Student, Hostel Attendant, Hostel Warden, and ADSA.

1. LEVEL 0: CONTEXT DIAGRAM (SYSTEM BOUNDARY)

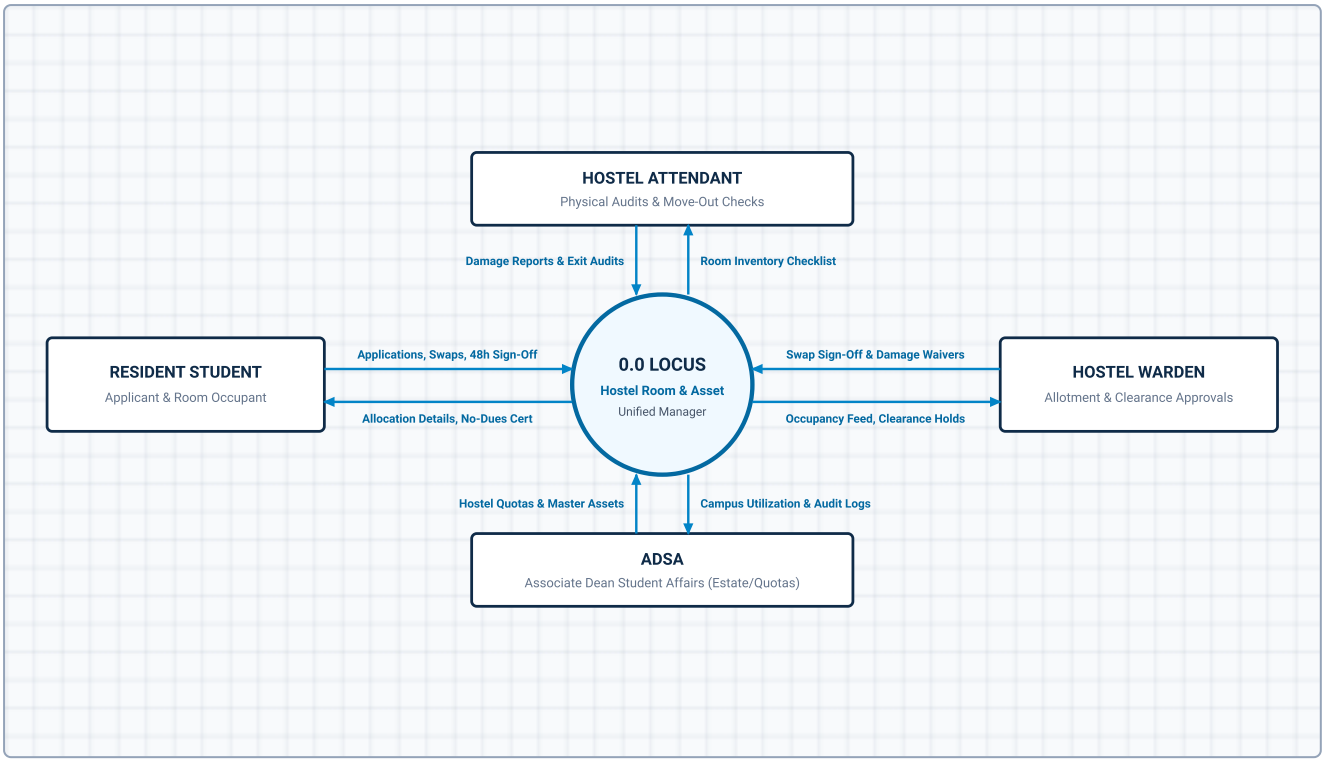


Figure 1.0: Level 0 DFD Context Diagram — Enforcing Boundary Abstraction without Internal Stores

Solid Border Box: External Entity | **Circle:** Root Process (0.0 LOCUS) | **Arrow:** External Data Flow | **Rule:** Databases Encapsulated Internally

2. DATA DICTIONARY: EXTERNAL ENTITY INTERFACES (LEVEL 0)

Entity Name	Data Inflow to Process 0.0	Data Outflow from Process 0.0	Operational Purpose
Resident Student	Room Application, Preferences, 48h Move-In Sign-off, Swap Request, Exit Form	Allotment Letter, Assigned Furniture Checklist, Digital No-Dues Certificate	Applies for housing, takes custody of room furniture, requests mutual swaps, and secures exit clearance.
Hostel Attendant	QR Audit Scans, Defect Reports, Room Move-Out Inspection Records	Live Room Asset Checklist, Audit Tasks, Discrepancy Alerts	Conducts physical room scans, confirms condition states, and checks baseline items during departures.
Hostel Warden	Allotment Confirmations, Mutual Swap Approvals, Damage Penalty/Waivers	Wing Occupancy Metrics, Clearance Hold Notifications, Dispute Escalations	Administrative oversight over room capacities, mutual swap authorizations, and asset dispute resolutions.
ADSA	Hostel Master Quotas, Gender/Branch Rules, Baseline Furniture Sets, Procurement Batches	Campus-wide Occupancy Analytics, Asset Health Summaries, Financial Damage Recoveries	Apex institutional authority configuring campus-wide accommodation policy and bulk furniture master stock.

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LEVEL 1: SYSTEM DECOMPOSITION WITH PERSISTENT DATABASES (D1–D5)

Document: LOCUS-DFD-01
Standard: Gane & Sarson

Level 1 decomposes Process 0.0 into four discrete subsystems and **formally unveils all five central databases (D1 through D5)**. Each persistent store maintains relational integrity across physical topology, student occupancies, barcode/QR asset registries, and immutable graduation clearance sign-offs.

1. LEVEL 1: SUBSYSTEM & DATABASE ARCHITECTURE DIAGRAM

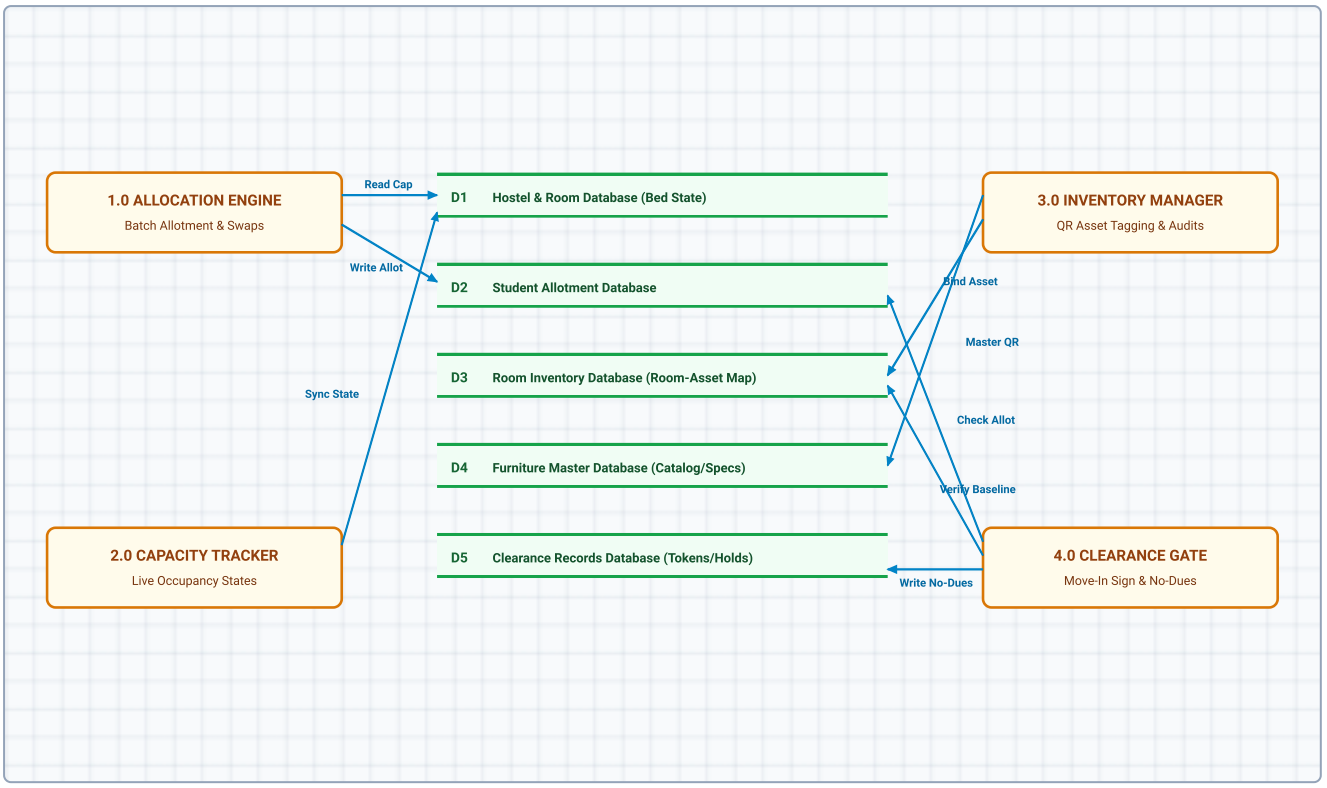


Figure 1.1: Level 1 DFD — Systematic Interplay Across Subsystems and the Five Persistent Databases

Rounded Box: Level 1 Subsystem **Parallel Bars:** Database Store (D1–D5) **Arrow:** Read / Write Operation

2. DATA DICTIONARY: PERSISTENT DATABASES (LEVEL 1)

ID	Database Name	Primary Key	Key Stored Attributes	Read / Write Subsystems
D1	Hostel & Room Database	`room_id`	hostel_id, block_id, room_no, bed_capacity, occupancy_state	Read/Write: 1.0, 2.0, 4.0
D2	Student Allotment Database	`allotment_id`	student_id, room_id, bed_id, session_year, allotment_status	Read/Write: 1.0; Read: 4.0
D3	Room Inventory Database	`inv_id`	room_id, asset_id, condition_status, last_audit_date	Read/Write: 3.0, 4.0
D4	Furniture Master Database	`asset_id`	item_type, qr_hash, procurement_batch, specs, standard_life	Write: 3.0 (ADSA Batch); Read: 3.0
D5	Clearance Records Database	`clearance_id`	student_id, movein_sign_ts, audit_sign_ts, damage_dues, no_dues_token	Read/Write: 4.0; Read: 1.0

Level 2.1 deconstructs **Process 1.0 (Student Allotment Engine)** into granular sub-processes. It explicitly traces incoming student preferences, quota validations, automated matching against available bed records in **D1**, allotment generation in **D2**, and mutual room exchange handling.

1. LEVEL 2.1: PROCESS 1.0 SUB-PROCESS FLOW DIAGRAM

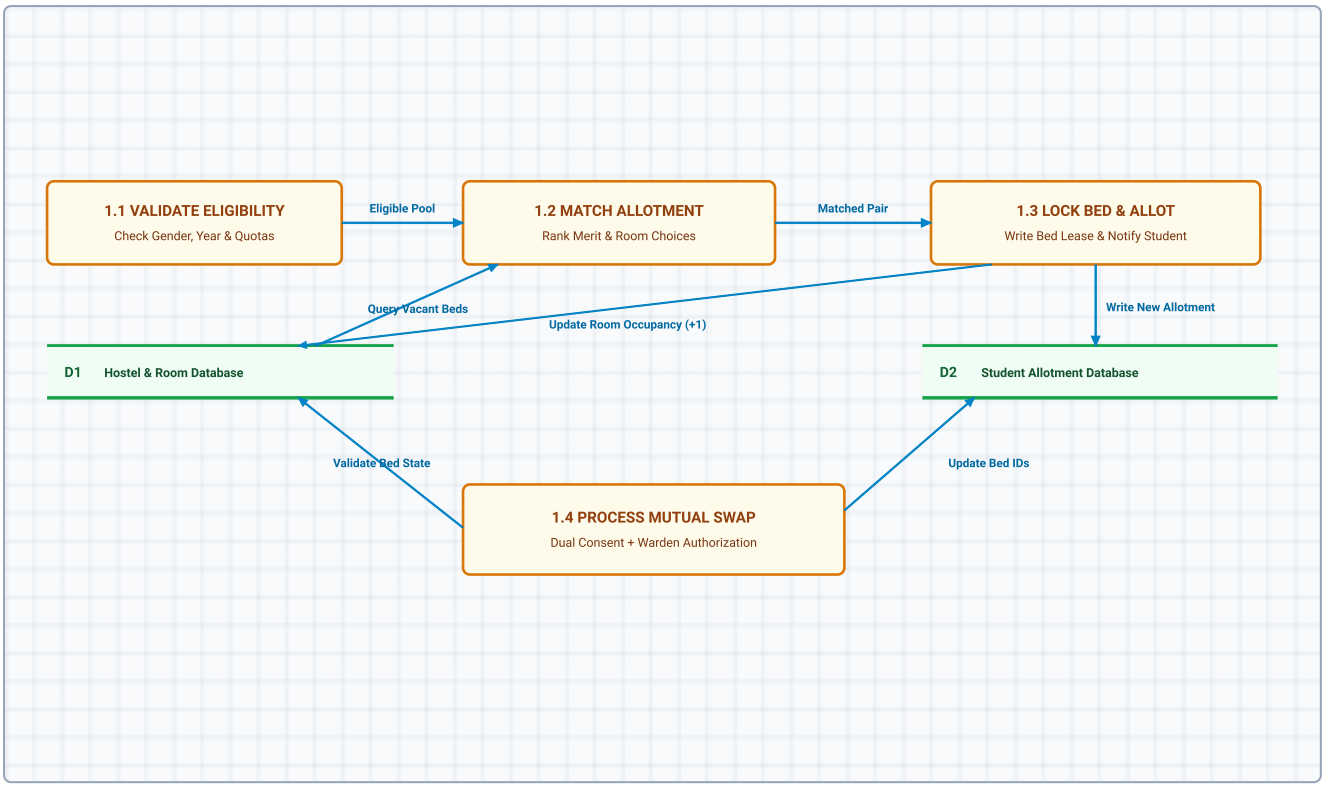


Figure 1.2: Level 2.1 DFD — Detailed Decomposition of Process 1.0 (Student Allotment & Transfers)

Sub-process: Operational step **Data Store:** D1 Room Store / D2 Allotment Store **Arrow:** Structured Data Movement

2. SUB-PROCESS LOGIC & DATA TRANSFORMATIONS (LEVEL 2.1)

Process #	Process Name	Input Flow & Origin	Output Flow & Destination
1.1	Validate Eligibility	Student Application Form (from Resident Student), Quotas & Branch Rules (from ADSA)	Categorized & verified candidate profile to Process 1.2; Rejection notice if ineligible.
1.2	Match Allotment	Verified candidate pool, Vacant bed records (read from **D1: Hostel & Room DB**)	Rank-matched student-bed assignment pair forward to Process 1.3.
1.3	Lock Bed & Allot	Matched assignment pair from Process 1.2	Writes lease record into **D2: Allotment DB**; updates occupancy count in **D1: Room DB**; sends Allotment Notice.
1.4	Process Mutual Swap	Dual Student Swap Requests, Warden Approval token; current lease records (from **D2**)	Re-assigns bed identifiers in **D2: Allotment DB**; updates occupancy tags in **D1: Room DB**.

Level 2.2 deconstructs **Processes 3.0 (Inventory Manager) and 4.0 (Clearance Gate)**. It models QR furniture binding, condition audits by Hostel Attendants, the 48-hour move-in student verification window, physical move-out inspections, and the automated release of the No-Dues token through **D3, D4, and D5**.

1. LEVEL 2.2: PROCESSES 3.0 & 4.0 SUB-PROCESS FLOW DIAGRAM

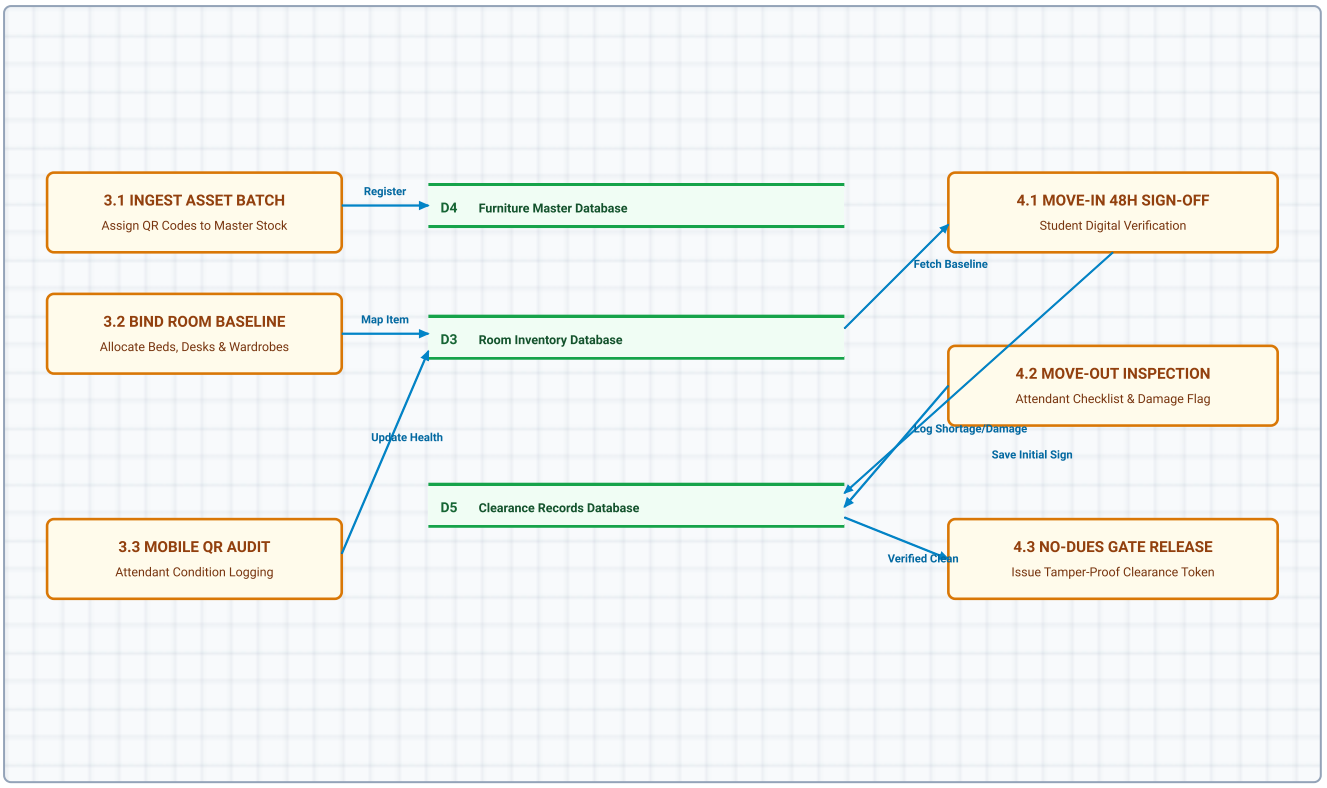


Figure 1.3: Level 2.2 DFD — Sub-Process Decomposition of Asset Management & Exit Clearance Gate

Sub-process: Operational step **Data Stores:** D3 Room Asset DB, D4 Furniture Master DB, D5 Clearance DB **Arrow:** Read / Write Stream

2. SUB-PROCESS LOGIC & CLEARANCE VERIFICATION RULES (LEVEL 2.2)

Sub-Process	Operational Trigger	Database Interaction (Read/Write)	Exit Gate Pass Condition
3.1 Ingest Batch	ADSA registers newly procured furniture lots	Writes new UID and QR hashes into **D4: Furniture Master DB**	Every physical asset has a unique scannable barcode/QR.
3.2 Bind Baseline	Caretaker outfits rooms per designated capacity	Maps item IDs to room ID in **D3: Room Inventory DB**	Total room assets match defined standard baseline set.
3.3 Mobile Audit	Attendant conducts door/furniture camera scan	Updates condition timestamp in **D3: Room Inventory DB**	Asset marked `Good` or `Fair`; discrepancies flagged.
4.1 Move-In Sign	Resident Student occupies room (48h limit)	Reads baseline from **D3**; records sign-off timestamp in **D5**	Student accepts room custody or notes pre-existing defects.
4.2 Move-Out Audit	Student initiates hostel checkout request	Attendant audits **D3** assets; logs unverified/damaged flags in **D5**	All assigned furniture items physically present and intact.
4.3 No-Dues Gate	Student requests final No-Dues certificate	Evaluates holds in **D5**; verifies Warden waiver if penalized	**ZERO pending holds in D5** triggers digital token generation.